

Computational Extension of OEIS Sequence A226188: Certified Inverse Harmonic Thresholds up to $n = 67$

Carlo Corti
Independent researcher
carlo.corti@outlook.com

June 26, 2026

Abstract

OEIS sequence A226188 is defined by

$$a(n) = \min\{k \geq 1 : H_k > 2n/3\},$$

where H_k is the k -th harmonic number. The local OEIS snapshot used for this computation contained 27 known terms. This paper reports a certified extension through $n = 67$. For every reported value $k = a(n)$, the certificate is the strict threshold condition

$$H_{k-1} \leq 2n/3 < H_k.$$

The computation uses the classical asymptotic approximation only to locate a small search neighborhood. The final decisions are made by outward-rounded MPFR/GMP interval evaluations, using direct interval summation for small arguments and Euler–Maclaurin formulae for large arguments, followed by an expanding bracket and binary search. The resulting 64-bit implementation ends at

$$a(67) = 14054172515346621401;$$

the next index requires a wider integer type. The new data also give a finite numerical consistency check, through the computed range, for the OEIS ratio behavior associated with the asymptotic limit $a(n+1)/a(n) \rightarrow \exp(2/3)$.

2020 Mathematics Subject Classification. Primary 11Y55; Secondary 11B83, 11A25.

Key words and phrases. OEIS, harmonic numbers, inverse harmonic thresholds, Euler–Maclaurin summation, interval arithmetic, computational number theory.

1 Introduction

The harmonic numbers

$$H_k = \sum_{j=1}^k \frac{1}{j}$$

form a strictly increasing sequence. OEIS A226188 asks for the first index at which this sum exceeds the arithmetic progression $2n/3$:

$$a(n) = \min\{k \geq 1 : H_k > 2n/3\}.$$

The entry previously listed the 27 values

$$1, 2, 4, 8, 16, 31, 60, 116, 227, 441, 859, 1674, 3260, 6349, \\ 12367, 24088, 46916, 91380, 177984, 346666, 675214, 1315136, \\ 2561536, 4989191, 9717617, 18927334, 36865412.$$

It also records the asymptotic consequence

$$\frac{a(n+1)}{a(n)} \rightarrow \exp(2/3) = 1.947734041054676 \dots,$$

attributed in the local snapshot to a comment by Jean-François Alcover. The same local snapshot places A226188 in a small family of harmonic-threshold OEIS sequences, including A226187 for the threshold $n/3$, A226189 for $\sqrt{n}$, and A226190 for $\log n$ [2].

The present computation extends the table by 40 further terms, from $n = 28$ through $n = 67$. It is a finite computational certification, not a new proof of an infinite theorem about all later indices. The asymptotic formula

$$a(n) \sim \exp(2n/3 - \gamma)$$

is used as a locator only. No reported value is accepted because it is close to the asymptotic estimate; each value is accepted only after the two neighboring harmonic inequalities have been certified.

The main contributions are:

1. reproduction of the 27 OEIS terms as a required gate;
2. certified generation of $a(n)$ for $28 \leq n \leq 67$;
3. a reproducible C17/MPFR/GMP implementation using an expanding bracket, binary search, and directed interval arithmetic;
4. an explicit 64-bit boundary: the current implementation supports $n \leq 67$, while $n = 68$ belongs to a wider-integer continuation.

2 Definition and Certification Predicate

Definition 1. Let $H_0 = 0$, and for $k \geq 1$ let

$$H_k = \sum_{j=1}^k \frac{1}{j}.$$

For $n \geq 1$, let

$$a(n) = \min\{k \geq 1 : H_k > 2n/3\}.$$

Because H_k is strictly increasing in k , the following equivalent predicate is the natural certificate for a reported term.

Proposition 2. For $n \geq 1$ and $k \geq 1$, $k = a(n)$ if and only if

$$H_{k-1} \leq \frac{2n}{3} < H_k.$$

Proof. If $k = a(n)$, then the defining strict inequality gives $H_k > 2n/3$, while minimality gives $H_{k-1} \leq 2n/3$. Conversely, if these two inequalities hold, then for every $j < k$ one has $H_j \leq H_{k-1} \leq 2n/3$, so k is the first index at which the harmonic sum exceeds the threshold. $\square$

This strict/weak pair is important: the OEIS definition uses $H_k > 2n/3$, so the lower neighbor must be allowed to equal the threshold.

3 Analytic Evaluation Framework

Direct summation to k is infeasible at the largest values in this paper. The final term is about $1.4 \cdot 10^{19}$, so a certification method must evaluate H_k without summing all reciprocals.

For large k , the implementation uses the Euler–Maclaurin expansion and the standard first-neglected-term remainder control for the logarithmic-gamma and psi-function asymptotics; see DLMF Section 5.11, especially the asymptotic material in Section 5.11(ii), in [5]. The Euler–Maclaurin intervals below are used only for $k > 10000$; smaller arguments are handled by direct interval summation. Here $\gamma = 0.5772156649 \dots$ is the Euler–Mascheroni constant and $\log$ denotes the natural logarithm. In the present integer setting $H_k = \psi(k+1) + \gamma$, where ψ is the digamma function, and the interval with four Bernoulli correction terms used for production comparisons is

$$H_k = \log k + \gamma + \frac{1}{2k} - \frac{1}{12k^2} + \frac{1}{120k^4} - \frac{1}{252k^6} + \frac{1}{240k^8} + R_4(k),$$

where $R_4(k)$ denotes the remainder after the displayed correction terms and

$$|R_4(k)| \leq \frac{1}{132k^{10}}.$$

This remainder estimate is the first-neglected-term bound from the cited DLMF asymptotic expansion. If $S_4(k)$ denotes the displayed sum without $R_4(k)$, then EM4 is the outward-rounded interval

$$\left[S_4(k) - \frac{1}{132k^{10}}, S_4(k) + \frac{1}{132k^{10}} \right].$$

As a higher-order check, the program also evaluates the interval with six Bernoulli correction terms

$$H_k = \log k + \gamma + \frac{1}{2k} - \frac{1}{12k^2} + \frac{1}{120k^4} - \frac{1}{252k^6} + \frac{1}{240k^8} - \frac{1}{132k^{10}} + \frac{691}{32760k^{12}} + R_6(k),$$

where $R_6(k)$ is the corresponding remainder after the displayed terms and

$$|R_6(k)| \leq \frac{1}{12k^{14}}.$$

This bound is used in the same way: if $S_6(k)$ denotes the displayed sum without $R_6(k)$, then EM6 is the outward-rounded interval

$$\left[S_6(k) - \frac{1}{12k^{14}}, S_6(k) + \frac{1}{12k^{14}} \right].$$

All endpoint operations are rounded outward with MPFR [6], linked against GMP [7]. Outward rounding means rounding lower endpoints toward $-\infty$ and upper endpoints toward $+\infty$. Thus each comparison uses an interval $[L_k, U_k]$ known to contain H_k . Here $[L_k, U_k]$ is the active direct-summation, EM4, or EM6 interval at the current working precision. The predicate $H_k > 2n/3$ is certified true only when $3L_k > 2n$, and certified false only when $3U_k \leq 2n$. Unresolved interval comparisons cause the program to increase precision or fail closed.

4 Sequence-Specific Structure and Cross-Checks

A226188 is not a rare-object search. Every positive n has exactly one answer, and the difficulty is to identify the crossing point of a monotone analytic function on a rapidly growing integer scale.

There is also a simple exact reduction on one residue class. If

$$b(t) = \min\{k \geq 1 : H_k > t\}$$

is the inverse harmonic threshold at an integer argument t , as in OEIS A002387, then

$$a(3m) = b(2m) \quad (m \geq 1),$$

because $2(3m)/3 = 2m$. Thus the rows with $n = 30, 33, \dots, 66$ in the new range coincide with the integer-threshold values $b(20), b(22), \dots, b(44)$, respectively. They are therefore external OEIS cross-check rows against A002387 rather than structurally new fractional-threshold cases. OEIS A004080 records the corresponding weak-threshold integer variant $H_k \geq t$, and the OEIS pages for these integer-threshold sequences cite the classical computation of Boas and Wrench on partial sums of the harmonic series [3, 4]. The remaining 27 new rows, where $n \not\equiv 0 \pmod{3}$, are the genuinely fractional-threshold cases for A226188 in this range.

The sequence-specific computational feature is that the asymptotic estimate is extremely accurate near the relevant k , but it is not itself a certificate. In the completed run, the seed

$$k_0 = \lfloor \exp(2n/3 - \gamma) \rfloor$$

was often equal or adjacent to the final value. Nevertheless, the algorithm does not use equality to the seed as an acceptance rule. It expands a bracket until it has a certified false endpoint and a certified true endpoint, and then uses the monotonicity of H_k to binary-search the first true value.

The certified ratios near the end of the new table are numerically consistent with the OEIS asymptotic comment. For example,

$$\frac{a(67)}{a(66)} \approx 1.947734041055,$$

and this quotient and $\exp(2/3)$ both round to 1.947734041055 at twelve decimal places. This is finite evidence in the certified range, while the limiting statement itself is supplied by the classical harmonic-number asymptotic expansion.

5 Algorithm and Implementation

The final implementation is the C17 source file `src/a226188.c`, built by `src/Makefile`. It has two supported modes:

- **verify-known**, which recomputes and checks the 27 OEIS terms;
- **generate**, which performs the known-term replay and then writes the certified range through $n = 67$.

For each n , the algorithm is:

1. compute the untrusted locator $k_0 = \lfloor \exp(2n/3 - \gamma) \rfloor$;

2. start with radius 16, set the trial lower endpoint to $\max(1, k_0 - r) - 1$, and set the trial upper endpoint to $k_0 + r$;
3. double the radius until the lower endpoint is certified false and the upper endpoint certified true for $H_k > 2n/3$, checking unsigned addition for overflow at each expansion and exiting without emitting a term if such an overflow is detected;
4. binary-search the resulting bracket while $k_{\text{high}} - k_{\text{low}} > 1$, using the overflow-safe midpoint $k_{\text{low}} + (k_{\text{high}} - k_{\text{low}})/2$;
5. certify the final neighbor inequalities $H_{k-1} \leq 2n/3 < H_k$;
6. repeat the final certificate with the higher-order EM6 interval at increased precision.

The loop invariant is that the lower endpoint is certified false and the upper endpoint is certified true. Since H_k is strictly increasing, the invariant proves that the final upper endpoint is the least true value. The bracket is finite, and each binary-search step strictly decreases $k_{\text{high}} - k_{\text{low}}$, so the search terminates. If an interval comparison is unresolved at 256 bits, the implementation retries at 384 and then 512 bits; if the comparison is still unresolved, it exits without emitting a term. In the completed run no escalation was needed beyond 256 bits.

For $k \leq 10000$, the program uses directed MPFR summation, the direct-sum instance of outward interval rounding: lower accumulators are formed with downward rounding and upper accumulators with upward rounding, so that the final pair encloses the exact finite sum. The threshold 10000 is a conservative implementation cutoff: direct summation is still inexpensive there, while Euler–Maclaurin evaluation is preferable on the much larger values occurring in the certified range. For larger values, the program uses the Euler–Maclaurin interval described above. The precision policy is 256, 384, and 512 MPFR bits; the completed run resolved all terms at 256 bits. The manifest records MPFR version 4.2.1 and GMP version 6.3.0.

6 Certification and Reproducibility

The production manifest `outputs/a226188_manifest.json` reports:

range	$1 \leq n \leq 67$,
known replay	$1 \leq n \leq 27$, pass,
new terms	$28 \leq n \leq 67$, certified,
terms written	67,
predicate evaluations	467,
elapsed time	0.027904 s,
maximum precision used	256 bits.

A local final verification run also recorded a wall time of about 0.03 s and a peak resident set size of about 3184 KB. The manifest timing is for the **generate** mode, which includes the known-term replay and generation through $n = 67$. These timing and memory figures are local telemetry, not a hardware-normalized benchmark. The current local verification environment is an x86-64 GNU/Linux system with an AMD Ryzen 9 7940HS CPU, 64 GB DDR5 installed RAM, GCC 13.3.0, and the Makefile flags `-std=c17 -O2` with warning flags `-Wall, -Wextra, -Wshadow, -Wconversion, -Wdouble-promotion, -Wformat=2`, and `-pedantic`.

The validation table `outputs/a226188_validation.tsv` contains one certificate row per index. Each row records the candidate k , seed, bracket endpoints, MPFR precision, harmonic method, two

comparison outcomes, positive margins for both neighboring inequalities, checker status, predicate evaluation count, elapsed time, and final status. All 67 rows have status **CERTIFIED** and checker status **PASS**. The smallest recorded margin for the lower-neighbor inequality is

$$2n/3 - U_{k-1} = 2.633584523432143 \cdot 10^{-22}$$

at $n = 67$, and the smallest recorded margin for the strict crossing inequality is

$$L_k - 2n/3 = 6.228932742447755 \cdot 10^{-21}$$

at $n = 65$. These margins are taken from the validation TSV and explain why the 256-bit interval comparisons were resolved in the completed run.

The file `data/b226188.txt` is the project’s canonical OEIS-ready local b-file; it contains 67 data rows and the final blank line required by the local workflow. Release-level file-integrity records are maintained with the repository/archive package rather than printed in the paper.

7 Results

Theorem 3 (Certified extension of A226188). *Subject to the recorded source, manifest, and validation table described above, the values in Table 1 are certified values of A226188 for $28 \leq n \leq 67$. Together with the reproduced OEIS terms for $1 \leq n \leq 27$, they give a certified table for $1 \leq n \leq 67$. For each listed value $k = a(n)$, the recorded certificate proves*

$$H_{k-1} \leq 2n/3 < H_k.$$

Computational proof. This is a finite computational certification conditioned on the recorded source, Makefile, manifest, and validation table. The public GitHub repository and Zenodo archive fix the source, build recipe, and released data artifacts. The program first recomputes $a(n)$ for $1 \leq n \leq 27$ and compares the results with the local OEIS b-file values. It then processes $n = 28, \dots, 67$ by certified bracketing and binary search. Proposition 2 shows that the final two inequalities for a row are sufficient to identify the least crossing index. The validation TSV records both inequalities and their positive margins for every n , and the manifest records completion of the run. $\square$

Given a printed pair (n, k) , the certificate for that individual row can be checked independently by evaluating H_{k-1} and H_k with any rigorous arbitrary-precision interval method and verifying the two inequalities in Proposition 2. The source and output artifacts are used to reproduce the generation and the exact table, not as a substitute for the mathematical certificate.

Corollary 4. *The local b-file `data/b226188.txt` extends the A226188 table to 67 indices. It contains the 27 previously known values followed by the 40 new certified terms in Table 1.*

8 Domain Limitations

The computation reported here is bounded by the integer domain of the final implementation. The largest certified value is

$$a(67) = 14054172515346621401,$$

which fits in `uint64_t`. The manifest records $n = 67$ as the maximum supported index for this implementation family and records $n = 68$ as requiring a wider integer type. This boundary is

Table 1: Certified terms of A226188 for $28 \leq n \leq 67$, extending the 27 previously known terms.

n	$a(n)$	n	$a(n)$
28	71804019	48	44334502845080
29	139855131	49	86351820384599
30	272400600	50	168190380070122
31	530563922	51	327590128640500
32	1033397411	52	638058445066582
33	2012783315	53	1242768153638596
34	3920366580	54	2420581837980561
35	7635831442	55	4714649644993433
36	14872568831	56	9182883605200051
37	28967808590	57	17885814992891026
38	56421586886	58	34836810713659947
39	109894245429	59	67852842108773716
40	214044762737	60	132159290357566703
41	416902270693	61	257411148671061652
42	812014744422	62	501368456813612867
43	1581588759549	63	976532410446924923
44	3080514265923	64	1902025418020652442
45	6000022499693	65	3704639653630074470
46	11686448069747	66	7215652763216299614
47	22762092724463	67	14054172515346621401

also visible directly from the harmonic threshold: an EM4 upper interval, evaluated with the same outward-rounded interval method used above, gives

$$H_{2^{64}-1} < 44.938636 < \frac{136}{3},$$

so no `uint64_t` value can satisfy the defining inequality for $n = 68$. Continuing the sequence should therefore be treated as a new wider-integer computation, not as a simple extension of the same compiled artifact.

The paper also does not claim an independent exhaustive reimplementaion of the full non-hit search. The validation artifacts certify the reported crossing values by interval arithmetic and replay the 27 known terms. The finite extension is reproducible from the stated source and output artifacts, but it is not a proof of all future terms.

9 Conclusion

OEIS A226188 has been extended from 27 known terms to a certified table through $n = 67$. The computation confirms every row by the defining inequalities

$$H_{a(n)-1} \leq 2n/3 < H_{a(n)}$$

and records the result in an OEIS-ready local b-file. The new values are finite certified data consistent, through the largest 64-bit term, with the ratio behavior predicted by the harmonic-number asymptotic expansion. Further extension begins at a genuine implementation boundary: $n = 68$ requires integers wider than `uint64_t`.

Data availability

The code, data files, result tables, manifest, and validation records are part of the public GitHub repository and Zenodo archive listed in the References. The essential local artifacts are `src/a226188.c`, `src/Makefile`, `data/b226188.txt`, `outputs/a226188_terms.tsv`, `outputs/a226188_validation.tsv`, `outputs/a226188_manifest.json`. The local b-file contains 67 data rows and ends with the required final blank line.

AI use disclosure

Large language models were used as research and drafting assistants during the A226188 workflow, including OpenAI Codex and external LLM-based review systems from the ChatGPT, Claude, DeepSeek, Gemini, Grok, Kimi, Qwen, and Z families. They are named at family level where exact model-version metadata were not consistently available in the local review artifacts.

Mathematical analysis. LLM assistance was used to organize the OEIS snapshot, identify the inverse-harmonic threshold structure, and check that the final predicate should be $H_{k-1} \leq 2n/3 < H_k$. The mathematical acceptance criterion in this paper is the explicit interval certificate recorded for each row, together with the author’s direct derivation and human-reviewed proof of Proposition 2.

Algorithm and code. LLM assistance was used in the design and review loop for the C17 implementation using MPFR and GMP. The final claims rely on the local source, manifest, validation TSV, known-term replay, and sanitizer/static-analysis records, not on conversational assertions.

Manuscript. LLM assistance was used to draft and structure this manuscript from the project artifacts. Numerical claims in the manuscript were checked against the local b-file, result tables, manifest, validation records, and final verification notes.

The author takes full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries, and conclusions of this paper.

References

- [1] OEIS Foundation Inc., *Sequence A226188: Least positive integer k such that $1 + 1/2 + \dots + 1/k > 2n/3$* , The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A226188>. Author-checked OEIS URL; local project snapshot used as source material for this manuscript.
- [2] OEIS Foundation Inc., *Related harmonic-threshold sequences A226187, A226189, and A226190*, The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A226187>, <https://oeis.org/A226189>, <https://oeis.org/A226190>. These related entries are cross-referenced in the local A226188 snapshot; the OEIS URLs were checked by the author.
- [3] OEIS Foundation Inc., *Sequences A002387 and A004080: integer-threshold inverse harmonic numbers*, The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A002387>, <https://oeis.org/A004080>. These OEIS sources provide the integer-threshold comparison used in the $a(3m) = b(2m)$ observation; the OEIS URLs were checked by the author.
- [4] R. P. Boas, Jr. and J. W. Wrench, Jr., *Partial sums of the harmonic series*, American Mathematical Monthly 78 (1971), no. 8, 864–870. <https://doi.org/10.1080/00029890.1971.11992881>.

- [5] NIST Digital Library of Mathematical Functions, *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>, Release 1.2.7 of 2026-06-15. Release date confirmed by the DLMF update note <https://dlmf.nist.gov/about/news/#S1.I1.i1>. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds. Section 5.11 was used for the asymptotic expansion and remainder discipline.
- [6] L. Fousse, G. Hanrot, V. Lefevre, P. Pelissier, and P. Zimmermann, *MPFR: A multiple-precision binary floating-point library with correct rounding*, ACM Transactions on Mathematical Software 33 (2007), no. 2, Article 13, 15 pp.
<https://doi.org/10.1145/1236463.1236468>.
- [7] GNU MP development team, *GNU MP: The GNU Multiple Precision Arithmetic Library, version 6.3.0*, <https://gmplib.org/manual/>.
- [8] C. Corti, *Computational artifacts for OEIS A226188*, GitHub repository <https://github.com/carcorti/A226188> and Zenodo archive, <https://doi.org/10.5281/zenodo.20960659>.